**CPPC 21/L**

**DIGITAL SIGNAL PROCESSING**

# ACTIVITY No.1

## Heuns Method, Impicit Euler Method, Runge Kutta 4th order method, and Wave Animation

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## I. Objectives

- To use MATLAB in applying numerical methods such as:
  - Heun's Method
  - Implicit Euler Method
  - Runge-Kutta 4th Order Method
- To understand how numerical methods approximate the solution of differential equations.
- To generate graphs that show the behavior of the computed results.
- To create and run a wave animation using MATLAB.
- To visualize mathematical and signal-related results through plots and animation.

## II. Materials

- MATLAB software
- Laptop

## III. Procedures

### 1. Open MATLAB

- Launch the MATLAB application.
- Click New Script to create a new coding file.
- Prepare separate scripts for each method to keep the activity organized.
- Save each script using the required filename before running it.

### 2. Run Heun's Method

- Open a new script in MATLAB.
- Type or paste the MATLAB code for Heun's Method.
- This method uses a predictor-corrector approach to approximate the solution.
- Save the file as:
  - LOGATOC_Act1_Heuns_Method.m
- Click Run to execute the script.
- A graph showing the Heun's Method solution should appear.

```
clc; clear;
```

```
h = 0.1;
t = 0:h:10;
n = length(t);

y = zeros(1,n);
v = zeros(1,n);

y(1) = 1;
v(1) = 0;

for i = 1:n-1
    y_predict = y(i) + h*v(i);
    v_predict = v(i) + h*(-y(i));

    y(i+1) = y(i) + h*(v(i) + v_predict)/2;
    v(i+1) = v(i) + h*(-y(i) - y_predict)/2;
end

plot(t,y)
title('Heun Method Solution')
xlabel('t')
ylabel('y(t)')
grid on
```

**3. Run Implicit Euler Method**

- Create another new script in MATLAB.
- Enter the code for the Implicit Euler Method.
- This method calculates the next value using an updated form of the derivative.
- Save the script as:
  - LOGATOC_Act1_Implicit_Euler_Method.m
- Click Run to display the graph of the solution.

```
clc; clear;

h = 0.1;
t = 0:h:10;
n = length(t);

y = zeros(1,n);
v = zeros(1,n);

y(1) = 1;
v(1) = 0;

for i = 1:n-1
    v(i+1) = (v(i) - h*y(i)) / (1 + h^2);
    y(i+1) = y(i) + h*v(i+1);
```

```
end

plot(t,y)
title('Implicit Euler Method Solution')
xlabel('t')
ylabel('y(t)')
grid on
```

**4. Run Runge-Kutta 4th Order Method**

- Open a new MATLAB script.
- Type or paste the code for the Runge-Kutta 4th Order Method.
- This method uses four slope values to produce a more accurate approximation.
- Save the file as:
  - LOGATOC_Act1_Runge_Kutta_4th_Order_Method.m
- Click Run to generate the RK4 graph.

```
clc; clear;

h = 0.1;
t = 0:h:10;
n = length(t);

y = zeros(1,n);
v = zeros(1,n);

y(1) = 1;
v(1) = 0;

for i = 1:n-1
    k1y = h*v(i);
    k1v = h*(-y(i));

    k2y = h*(v(i) + k1v/2);
    k2v = h*(-(y(i) + k1y/2));

    k3y = h*(v(i) + k2v/2);
    k3v = h*(-(y(i) + k2y/2));

    k4y = h*(v(i) + k3v);
    k4v = h*(-(y(i) + k3y));

    y(i+1) = y(i) + (k1y + 2*k2y + 2*k3y + k4y)/6;
    v(i+1) = v(i) + (k1v + 2*k2v + 2*k3v + k4v)/6;
end

plot(t,y)
title('RK4 Solution of Second-Order Differential Equation')
xlabel('t')
ylabel('y(t)')
```

```
grid on
```

**5. Run Wave Animation**

- Create another new script for the wave animation.
- Type or paste the MATLAB code for the 3D wave propagation.
- The animation uses a sine function to show the movement of a wave.
- Save the file as:
  - LOGATOC_Act1_wave_animation.m
- Click Run to view the animated wave surface.

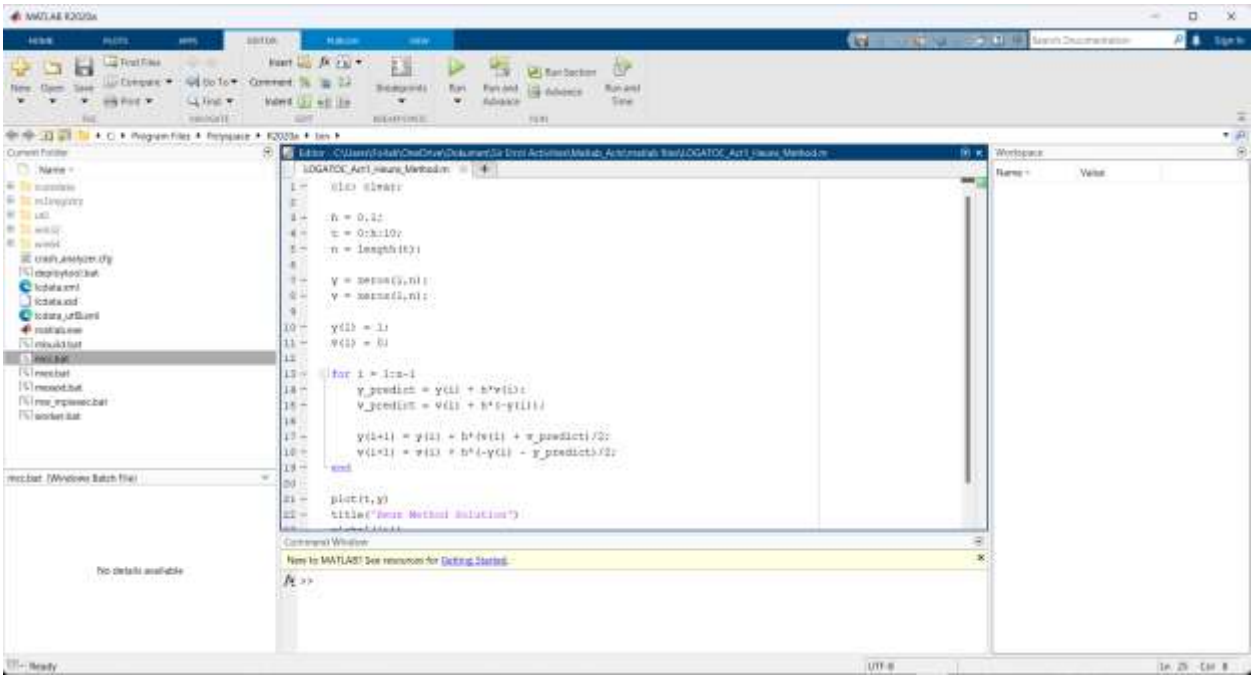
```
clc; clear;

[x,t] = meshgrid(0:0.2:10, 0:0.2:10);

for k = 1:50
    z = sin(x - t + k*0.1);

    surf(x,t,z)
    xlabel('X')
    ylabel('Time')
    zlabel('Amplitude')
    title('3D Wave Propagation')
    axis([0 10 0 10 -1 1])
    pause(0.1)
end
```

## IV. Conclusion

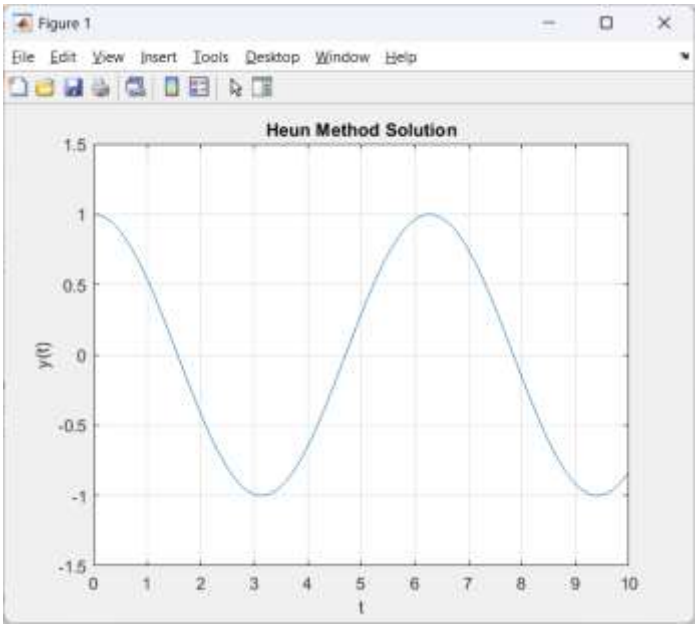
The activity showed how MATLAB can be used to apply different numerical methods and visualize their results through graphs and animation. By running the scripts for Heun's Method, Implicit Euler Method, and Runge-Kutta 4th Order Method, the approximate solutions were displayed and compared through plotted outputs. The wave animation also demonstrated how MATLAB can represent signal behavior in a visual and interactive way. Overall, this activity helped in understanding how numerical techniques and simulations can be implemented using MATLAB.

## V. Documentation

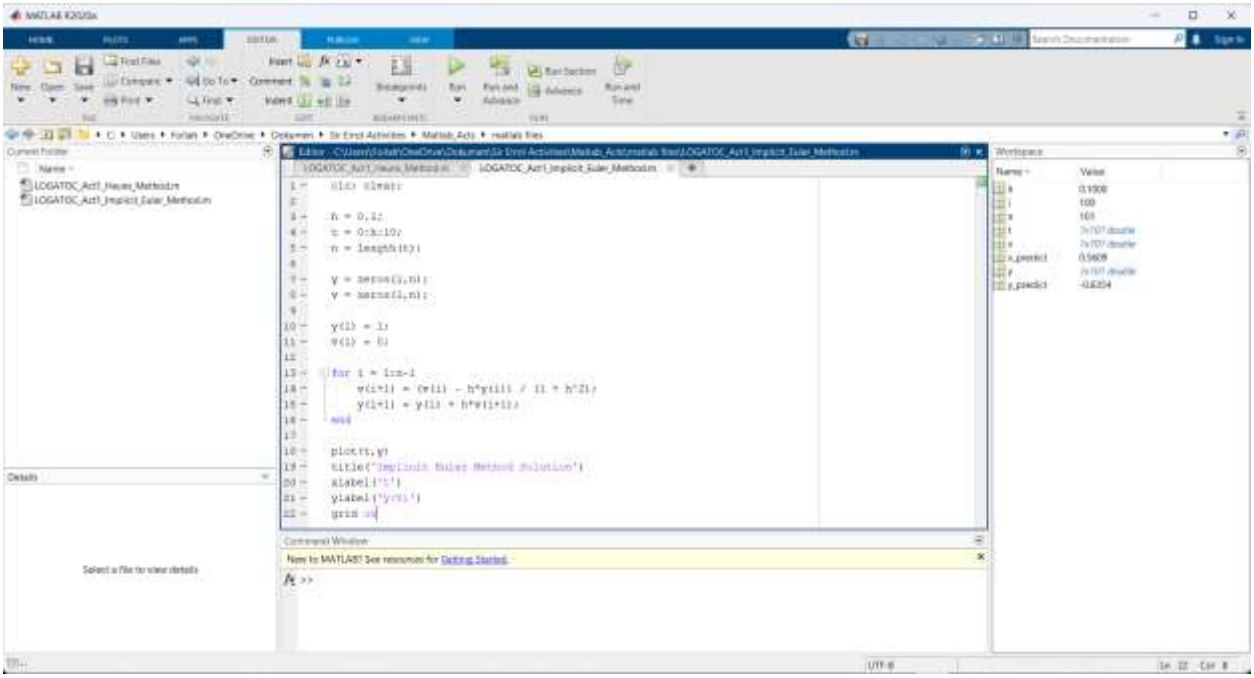
### Heun's Method



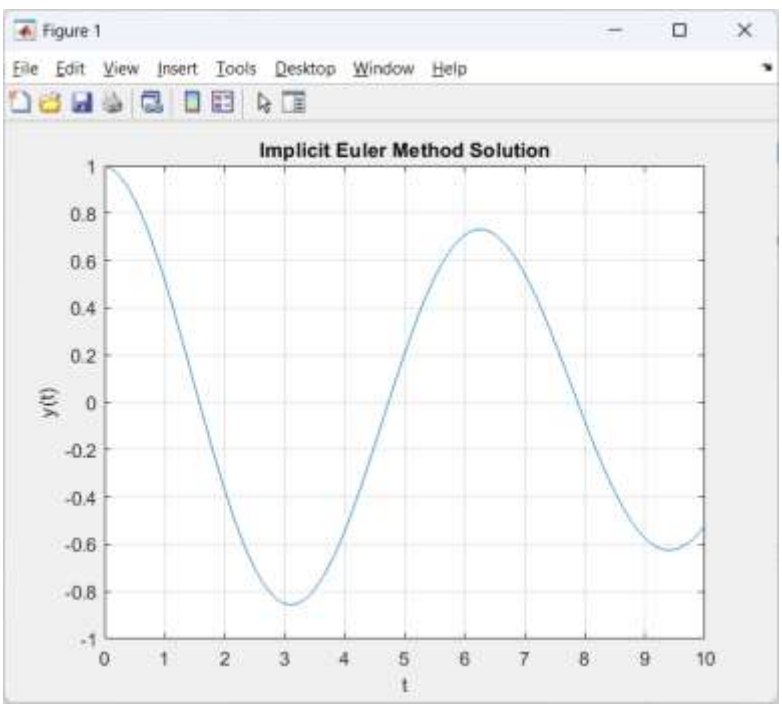

### Result:



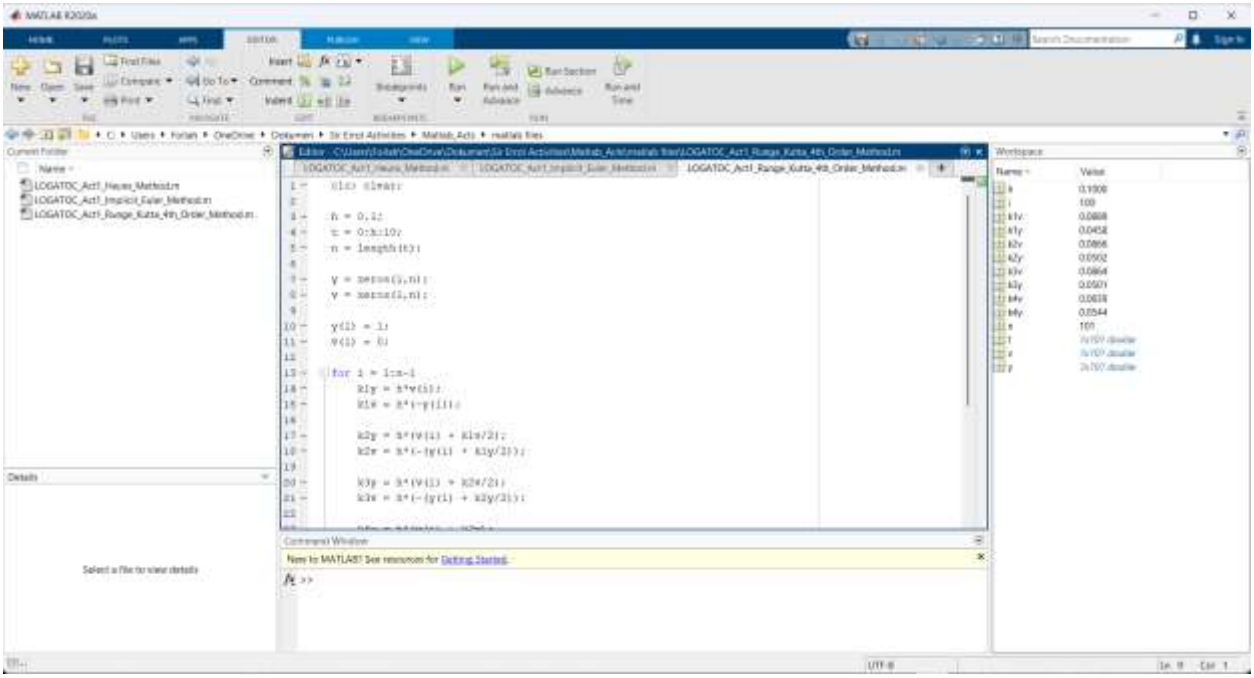
## Implicit Euler Method



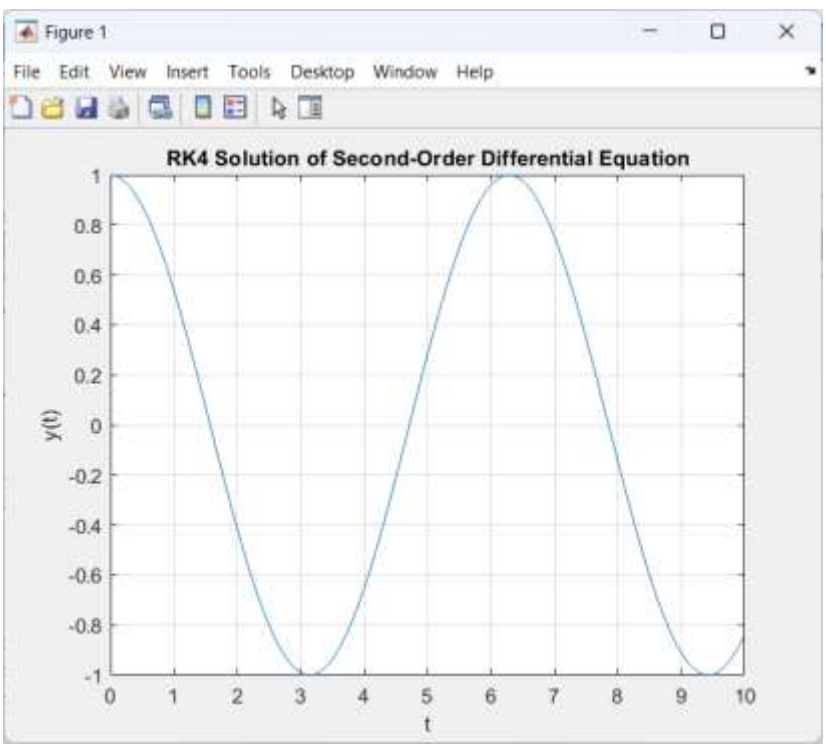

## Result:



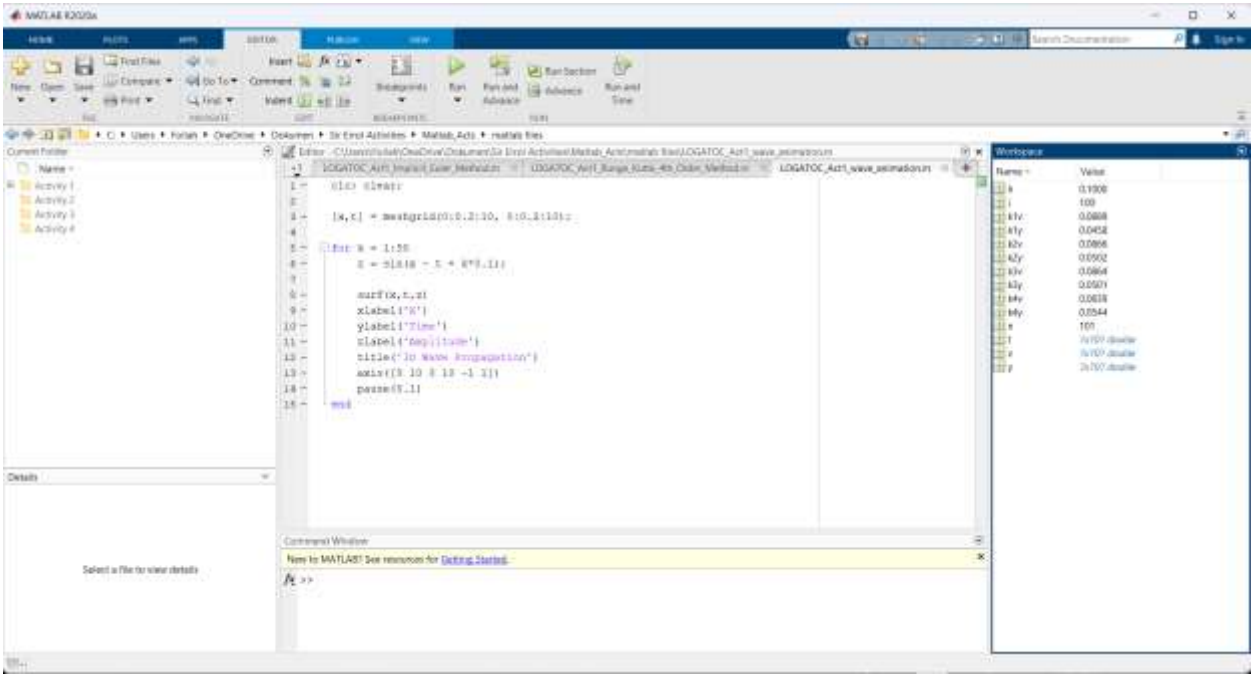
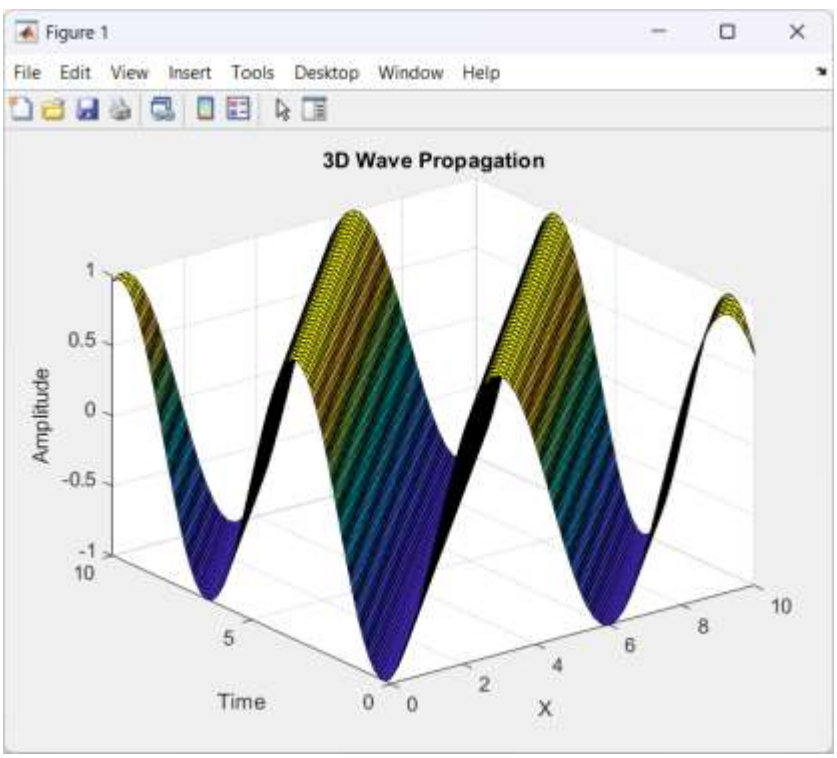
## Runge-Kutta 4th Order Method




## Result:



## Wave Animation



## Result: